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RESEARCH:

Virtual Try-On Pipelines

Introduction

Virtual Try-On (VTON) pipelines are Artificial Intelligence (AI) and Computer Vision based systems that allow users to digitally visualize how clothing items appear on their body without physically wearing them. These systems are widely used in fashion e-commerce, smart retail applications, and online shopping platforms to enhance customer experience and reduce product return rates. Virtual try-on technology combines image processing, human body analysis, segmentation, and deep learning techniques to generate realistic outputs in which users appear to wear selected garments. With the growth of online shopping and digital fashion technologies, virtual try-on systems have become an important research area in computer vision.

Input Acquisition and Preprocessing

The first stage of a virtual try-on pipeline involves collecting the required inputs. Generally, two inputs are used: a person image and a garment image. The person image may be captured through webcams, mobile cameras, or uploaded photographs, while the garment image is obtained from fashion catalogs or clothing datasets. Before processing, images undergo preprocessing operations such as resizing, normalization, background removal, and noise reduction. These operations improve image quality and prepare the data for later stages of the pipeline.

Human Parsing and Segmentation

Human parsing is an important stage in virtual try-on systems because it identifies and separates different body regions such as the face, hair, torso, arms, legs, and existing clothing. This process helps the system understand where the new garment should be placed. Clothing segmentation is also performed to detect upper wear, lower wear, sleeves, collars, and garment boundaries. Models such as U-Net, DeepLabV3, Mask R-CNN, and MediaPipe segmentation are commonly used for this purpose. Accurate segmentation improves cloth placement and reduces visual errors in the final output.

Pose Estimation

Pose estimation is used to detect body landmarks and understand the posture of the user. It identifies key points such as shoulders, elbows, wrists, hips, knees, and ankles. The extracted pose information helps align the selected garment with the user's body orientation. Popular frameworks used for pose estimation include OpenPose, MediaPipe Pose, and DensePose. This stage is especially important in real-time webcam-based systems because clothing must adapt to body movements and changes in posture.

Cloth Warping and Alignment

Cloth warping is considered the core stage of the virtual try-on pipeline. Since the shape of the garment image may differ from the user's posture, transformation techniques are applied to align clothing with the body structure. Methods such as Thin Plate Spline (TPS), Geometric Matching Networks (GMN), and Spatial Transformer Networks (STN) are commonly used. These methods preserve garment texture, logos, patterns, and boundaries while adjusting clothing size and orientation.

Image Generation and Output Rendering

The final stage generates the virtual try-on output by combining the aligned garment with the user image. Earlier systems used Generative Adversarial Networks (GANs) such as VITON and CP-VTON, while modern approaches increasingly use diffusion models due to their improved realism and texture preservation. Rendering

techniques such as edge smoothing, wrinkle generation, and color correction are applied to enhance visual quality.

Conclusion

Virtual try-on pipelines integrate segmentation, pose estimation, cloth alignment, and image generation techniques to create realistic clothing visualizations. Modern AI-based approaches continue to improve the accuracy and quality of virtual fitting systems, making them valuable tools in digital fashion and e-commerce applications.

AI Fitting Systems

Introduction

AI fitting systems are intelligent technologies that use Artificial Intelligence (AI), Computer Vision, and Machine Learning techniques to help users virtually try and fit clothing without physically wearing them. These systems analyze body shape, measurements, posture, and garment properties to provide personalized fitting experiences. AI fitting systems have become increasingly important in the fashion and e-commerce industries because they improve customer satisfaction, reduce return rates, and enhance online shopping experiences. With the growth of digital retail platforms, AI fitting technologies are being widely adopted in smart mirrors, virtual dressing rooms, and mobile shopping applications.

Working Principle of AI Fitting Systems

AI fitting systems operate by collecting user information and processing it through various AI models. The system generally takes inputs such as user images, videos, body measurements, and selected garment images. Computer vision algorithms then analyze body dimensions and detect important regions including shoulders, waist, hips, arms, and legs. The extracted information is used to determine how the garment fits on the user. The system finally generates a visualization showing the user wearing the selected outfit.

Body Measurement and Detection

One of the main components of AI fitting systems is body measurement estimation. Traditional fitting methods require manual measurements, whereas AI systems automatically estimate dimensions using cameras and image processing techniques. Technologies such as MediaPipe Pose, OpenPose, and DensePose are used to detect body landmarks and calculate measurements. Parameters including height, chest size, waist, hip ratio, and body posture are analyzed to provide accurate fitting recommendations. This process helps users select suitable clothing sizes.

Virtual Try-On and Garment Simulation

AI fitting systems commonly integrate virtual try-on technology to display clothing digitally on users. Garment simulation methods align and adapt clothes according to body posture and size. Techniques such as segmentation, cloth warping, and image generation are used to create realistic outputs. Modern systems also simulate fabric behavior, folds, and garment movement to improve realism. These features help users visualize how clothes look before purchasing.

Applications of AI Fitting Systems

AI fitting systems are widely used in fashion e-commerce platforms, smart mirrors, retail stores, and augmented reality applications. They support personalized shopping experiences and reduce uncertainty during online purchases. Fashion brands and retailers use these systems to increase customer engagement and minimize product returns caused by incorrect sizing.

Conclusion

AI fitting systems combine AI, computer vision, body analysis, and virtual try-on technologies to provide intelligent clothing fitting experiences. By improving size prediction, garment visualization, and personalization, these systems are transforming modern fashion retail and digital shopping environments.

Tasks:

Align Clothes on Body/Avatar

Introduction

Aligning clothes on the body or digital avatar is one of the fundamental tasks in AI fitting systems and virtual try-on technologies. The main objective of this process is to correctly place the selected garment on the user according to body structure, posture, orientation, and dimensions. Since garment images obtained from datasets, online stores, or fashion catalogs are usually available as flat images, they cannot be directly placed onto the user image. Therefore, alignment techniques are necessary to transform and position garments before generating the final virtual fitting output.

Body Analysis and Landmark Detection

The alignment process begins with analyzing the body structure of the user using computer vision techniques. AI systems identify important body landmarks such as shoulders, elbows, wrists, chest region, waist, hips, knees, and ankles. These landmarks help the system understand body posture and determine where the garment should be positioned.

Pose estimation models such as MediaPipe Pose, OpenPose, and DensePose are commonly used to extract these body points. For example, while fitting upper garments like shirts and jackets, shoulder width and torso position become important parameters. Similarly, lower garments such as trousers and skirts require accurate hip and leg alignment.

The detected body landmarks act as reference points that guide the clothing placement process and improve fitting accuracy.

Avatar-Based Clothing Alignment

Modern AI fitting systems increasingly use digital avatars for clothing alignment. A digital avatar is a virtual representation of the user created using body measurements, images, or 3D scans. Instead of directly placing garments on user images, clothing is first aligned with the avatar surface.

This method improves realism because garments adapt according to the avatar's dimensions and body posture. Avatar-based alignment is widely used in smart mirrors, virtual dressing rooms, gaming environments, and metaverse fashion applications.

The use of avatars also supports clothing visualization from different viewing angles and allows users to interact with garments more effectively.

Importance of Proper Alignment

Correct clothing alignment is important because improper placement can lead to unrealistic fitting results, distorted garments, and poor visualization quality. Misalignment may cause sleeve mismatches, incorrect collar positions, and improper waist fitting.

Therefore, accurate body analysis and alignment methods are essential to achieve realistic virtual try-on experiences.

Fit Clothes Realistically

Introduction

After aligning garments with the body or avatar, the next important task is realistic cloth fitting. The objective of realistic fitting is to make the garment appear naturally worn by the user instead of simply overlaying it on the image. The garment should behave as it would in real life while preserving its visual characteristics.

Realistic fitting significantly improves customer confidence and enhances online shopping experiences because users can understand how the clothing may appear before purchase.

Body Measurement Adaptation

AI fitting systems analyze body measurements such as height, shoulder width, chest dimensions, waist size, hip measurements, and body posture. These measurements help the system modify garment dimensions according to the user's body shape.

For example, if a user has broader shoulders or a different body proportion, the garment size and orientation are adjusted accordingly. This process ensures that the clothing adapts naturally instead of appearing stretched or compressed.

Automatic body measurement estimation reduces the need for manual size selection and improves fitting recommendations.

Preservation of Garment Appearance

One of the major requirements of realistic fitting is maintaining the original appearance of the garment. Important details such as logos, patterns, textures, embroidery, collars, sleeves, and printed designs must remain unchanged after fitting.

If these details become distorted during processing, the output loses realism. Therefore, AI systems focus on preserving garment characteristics while adapting the clothing to body dimensions.

Texture preservation techniques are especially important in garments containing stripes, logos, and complex designs.

Visual Enhancement and Cloth Simulation

Real clothes naturally create folds, wrinkles, and shadows depending on body posture and movement. AI fitting systems simulate these effects to improve realism.

Wrinkle simulation helps garments appear naturally fitted rather than artificially attached to the body. Shadow generation improves depth perception and makes clothing appear integrated with the user image.

Additional enhancement techniques include edge refinement, lighting correction, and color adjustment. These processes remove artifacts and improve visual consistency.

Modern systems often use Generative Adversarial Networks (GANs) and diffusion models to create highly realistic outputs with better texture preservation and cloth behavior.

Importance of Realistic Fitting

Realistic fitting improves user satisfaction and reduces uncertainty during online purchases. It allows customers to understand garment appearance and fitting quality before buying, thereby reducing product return rates.

Study Garment Warping

Introduction

Garment warping is considered the core operation in AI fitting systems because it transforms clothing images according to body posture and shape. Garment images are usually available in fixed and flat orientations, whereas users may appear in different poses and positions. Therefore, clothing must be reshaped before fitting.

The objective of garment warping is to adapt clothing geometry while preserving important garment features.

Purpose of Garment Warping

Garment warping modifies the shape and orientation of clothing to match the body structure of the user. During this process, important elements such as textures, logos, sleeves, collars, boundaries, and patterns must remain unchanged.

Effective warping ensures smooth adaptation of clothing while reducing distortions and visual artifacts.

The process generally follows:

Original Garment



Transformation



Warped Garment



Final Fitting

Warping Techniques

Several techniques are used in garment warping systems.

Thin Plate Spline (TPS) is one of the most widely used methods because it performs smooth transformations while maintaining garment structure. TPS adjusts garment control points according to body landmarks and produces flexible deformation.

Geometric Matching Networks (GMN) use deep learning methods to automatically learn transformations between body images and clothing images. These models improve fitting accuracy and reduce distortions.

Spatial Transformer Networks (STN) allow neural networks to learn image transformations automatically during training. STN improves cloth adaptation by handling scaling, rotation, and image deformation.

These methods help maintain garment quality while adapting clothing to complex body postures.

Challenges in Garment Warping

Garment warping faces several difficulties such as texture distortion, boundary artifacts, sleeve mismatches, shape deformation errors, and fitting inconsistencies.

Complex body poses further increase the difficulty because garments must adapt while preserving realism. Research continues to focus on improving warping methods to overcome these challenges and enhance fitting quality.

Importance of Garment Warping

Garment warping directly influences fitting accuracy and realism. Effective warping methods create natural clothing adaptation and improve the quality of virtual try-on systems.

Conclusion

Aligning clothes on body avatars, fitting garments realistically, and studying garment warping are the key tasks in AI fitting systems and virtual try-on applications. Clothing alignment ensures proper placement, realistic fitting improves visual appearance, and garment warping adapts clothing according to body posture and dimensions. Together, these processes enable accurate, realistic, and interactive virtual fitting experiences used in modern fashion and e-commerce technologies.

Try-On Demo Study

Introduction

Virtual Try-On (VTON) demo systems are developed to digitally visualize clothing on users without physically wearing garments. These systems use computer vision, image processing, and deep learning techniques to generate realistic clothing visualizations. Over time, several try-on models have been introduced to improve garment alignment, texture preservation, fitting quality, and image realism. Studying existing try-on demos helps in understanding the evolution of virtual fitting technologies and identifying techniques used in modern AI fitting systems.

This study focuses on popular virtual try-on models including **VITON**, **CP-VTON**, **HR-VITON**, and **LaDI-VTON**.

VITON (Virtual Try-On Network)

VITON is one of the earliest and most widely known virtual try-on systems. It introduced a framework for generating images where a person appears to wear a selected garment. The model takes two inputs: a person image and a clothing image. The garment is aligned and combined with the user image to produce the final output.

The VITON pipeline follows the workflow:

Person Image
+
Garment Image
↓
Clothing Alignment
↓
Image Generation
↓
Virtual Try-On Output

VITON mainly focuses on transferring garments while maintaining body posture and preserving user identity. It introduced the concept of clothing alignment and image synthesis for virtual fitting applications.

Features of VITON

- Supports image-based virtual try-on
- Uses garment transfer techniques
- Preserves body posture information
- Generates clothing visualization outputs

Limitations of VITON

Although VITON introduced virtual try-on concepts, it showed limitations such as texture distortion, alignment errors, and lower image quality in complex poses.

CP-VTON (Characteristic Preserving Virtual Try-On Network)

CP-VTON was developed as an improvement over VITON. The main objective of CP-VTON is to preserve garment characteristics while transferring clothing to the user image.

CP-VTON introduces geometric matching techniques to improve cloth alignment and reduce texture distortion.

Workflow:

Person Image
+
Garment Image
↓
Geometric Matching
↓
Cloth Warping
↓
Image Synthesis
↓
Output

CP-VTON improves the preservation of clothing patterns, logos, textures, and garment details during fitting.

Features of CP-VTON

- Better texture preservation
- Improved garment alignment
- Uses cloth warping methods
- Reduces visual distortions

Compared with VITON, CP-VTON produces more realistic clothing appearance and better garment adaptation.

HR-VITON (High Resolution Virtual Try-On Network)

HR-VITON was introduced to improve image quality and resolution in virtual try-on systems. Earlier methods produced low-resolution outputs with artifacts around garment boundaries. HR-VITON addresses these issues by generating high-quality try-on images.

Pipeline:

Person Image
+
Garment Image
↓
Alignment Module
↓
High Resolution Generation
↓
Final Output

The model improves cloth details, edge quality, and fitting realism.

Features of HR-VITON

- High-resolution output generation
- Improved texture quality
- Better garment boundary preservation
- Reduced artifacts and distortions

HR-VITON produces visually enhanced outputs suitable for advanced fashion applications.

LaDI-VTON (Latent Diffusion Virtual Try-On)

LaDI-VTON represents modern virtual try-on approaches based on diffusion models. Unlike traditional GAN-based methods, LaDI-VTON uses latent diffusion techniques to generate realistic clothing images.

Pipeline:

Person Image
+
Garment Image
↓
Condition Encoding
↓
Latent Diffusion Model



Generated Output

Diffusion-based methods improve image realism and texture preservation while reducing fitting errors.

Features of LaDI-VTON

- Diffusion-based architecture
- High realism output generation
- Better cloth adaptation
- Improved texture preservation
- Reduced visual artifacts

LaDI-VTON is considered one of the advanced virtual try-on approaches used in modern AI fashion systems.

Observations

The study shows that virtual try-on systems evolved from simple garment transfer methods to advanced AI-based fitting approaches. Early models such as VITON focused on garment placement, while later methods improved texture preservation and fitting quality. Modern diffusion-based systems such as LaDI-VTON provide realistic clothing generation and improved visual appearance.

Conclusion

The study of try-on demos provides understanding of virtual fitting technologies and their evolution. VITON introduced the concept of image-based virtual try-on, CP-VTON improved garment preservation, HR-VITON enhanced image quality, and LaDI-VTON introduced diffusion-based fitting methods. These systems form the foundation of modern AI fitting and virtual try-on applications used in fashion and e-commerce industries.

Work-flow:



Basic Virtual Try-On Experiment: Shirt Overlay on Detected Body

Objective

The objective of this experiment is to develop a simple Virtual Try-On prototype that overlays a shirt representation on a detected human body using computer vision techniques. The experiment aims to understand the basic workflow involved in virtual try-on systems, including body detection, alignment, garment placement, and visualization. This experiment serves as an introductory implementation for studying AI-based fitting systems and garment alignment techniques.

Methodology

The experiment was implemented using **OpenCV** and **MediaPipe Pose** libraries. The webcam was used to capture real-time video input from the user. MediaPipe Pose was employed to detect body landmarks such as shoulders and hips, which are important reference points for garment placement.

The system continuously captured video frames and analyzed the body posture of the user. Based on the detected shoulder and hip coordinates, the approximate torso region was estimated. A shirt representation was then generated and overlaid onto the detected body region.

Manual alignment functionality was also introduced to simulate clothing adjustment. Keyboard controls (**W, A, S, D**) were used to move the overlaid shirt in different directions to study garment positioning and alignment behavior.

Workflow of the Experiment

The experiment followed the workflow shown below:

```
graph TD; A[Input from Webcam] --> B[Body Detection using MediaPipe Pose]; B --> C[Landmark Extraction]; C --> D[Torso Region Estimation]; D --> E[Shirt Overlay]; E --> F[Manual Alignment]; F --> G[Final Visualization];
```

The implementation also conceptually connected previous modules:

```
graph TD; H[Dataset Preparation] --> I[Segmentation]; I --> J[AI Fitting Concepts]; J --> K[Virtual Try-On Visualization];
```

```

virtual.py > ...
1  import cv2
2  import mediapipe as mp
3  import numpy as np
4  mp_pose = mp.solutions.pose
5  pose = mp_pose.Pose()
6  cap = cv2.VideoCapture(0)
7  offset_x = 0
8  offset_y = 0
9  while True:
10     ret, frame = cap.read()
11     if not ret:
12         break
13     frame = cv2.flip(frame,1)
14     rgb = cv2.cvtColor(
15         frame,
16         cv2.COLOR_BGR2RGB
17     )
18     results = pose.process(rgb)
19     h,w,_ = frame.shape
20     cv2.putText(
21         frame,
22         "M1 -> M2 -> M3 -> M4",
23         (20,30),
24         cv2.FONT_HERSHEY_SIMPLEX,
25         0.8,
26         (0,255,255),
27         2
28     )
29     if results.pose_landmarks:
30         lm = results.pose_landmarks.landmark
31         ls = lm[11]
32         rs = lm[12]
33         lh = lm[23]
34         rh = lm[24]
35         lsx,lsy = int(ls.x*w),int(ls.y*h)
36         rsx,rsy = int(rs.x*w),int(rs.y*h)
37         lhx,lhy = int(lh.x*w),int(lh.y*h)
38         rhx,rhy = int(rh.x*w),int(rh.y*h)
39         shirt_w = abs(rsx-lsx)+120
40         shirt_h = abs(lhy-lsy)+120
41         x = min(lsx,rsx)-60+offset_x
42         y = min(lsy,rsy)+offset_y
43         x2 = x+shirt_w
44         y2 = y+shirt_h
45         cv2.rectangle(
46             frame,
47             (x,y),
48             (x2,y2),
49             (255,0,0),
50             -1

```

```

virtual.py > ...
52         cv2.rectangle(
53             frame,
54             (x-40,y+20),
55             (x,y+80),
56             (255,0,0),
57             -1
58         )
59         cv2.rectangle(
60             frame,
61             (x2,y+20),
62             (x2+40,y+80),
63             (255,0,0),
64             -1
65         )
66         cv2.circle(
67             frame,
68             (
69                 (x+x2)//2,
70                 y+20
71             ),
72             25,
73             (0,0,0),
74             -1
75         )
76         cv2.putText(
77             frame,
78             "SHIRT",
79             (
80                 x+40,
81                 y+shirt_h//2
82             ),
83             cv2.FONT_HERSHEY_SIMPLEX,
84             1,
85             (255,255,255),
86             3
87         )
88         mp.solutions.drawing_utils.draw_landmarks(
89             frame,
90             results.pose_landmarks,
91             mp_pose.POSE_CONNECTIONS
92         )
93         cv2.putText(
94             frame,
95             "WASD move shirt | ESC exit",
96             (20,h-20),
97             cv2.FONT_HERSHEY_SIMPLEX,
98             0.7,
99             (0,255,0),
100             2
101         )

```

```

99         (0,255,0),
100         2
101     )
102     cv2.imshow(
103         "Virtual Try-On Demo",
104         frame
105     )
106     key=cv2.waitKey(1)
107     if key==ord('a'):
108         offset_x-=10
109     elif key==ord('d'):
110         offset_x+=10
111     elif key==ord('w'):
112         offset_y-=10
113     elif key==ord('s'):
114         offset_y+=10
115     elif key==27:
116         break
117 cap.release()
118 cv2.destroyAllWindows()

```

Implementation Details

The prototype detects body landmarks in real time and dynamically adjusts the shirt placement according to body movement. The shirt dimensions are estimated using the distance between shoulder landmarks and torso height measurements.

The following functionalities were implemented:

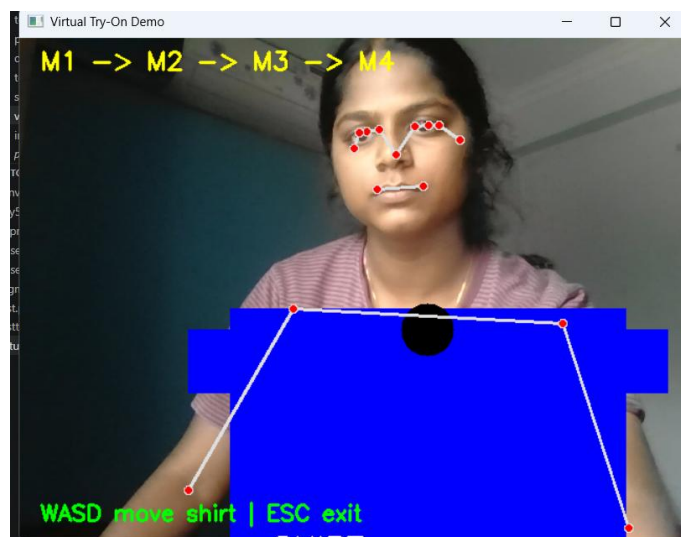
- Real-time webcam input acquisition
- Body pose detection using MediaPipe
- Shoulder and hip landmark extraction
- Dynamic shirt placement on torso region
- Manual alignment controls for garment adjustment
- Visualization of module workflow concepts

The system displayed the body skeleton together with the overlaid shirt to observe alignment behavior.

Observations

The experiment successfully demonstrated the basic operation of a virtual try-on pipeline. The shirt overlay followed body movements and adapted according to torso dimensions. Manual adjustment controls helped study clothing positioning and alignment issues.

Some limitations were observed during testing. Since a simple shirt representation was used instead of an actual garment image, realistic fitting effects such as cloth folds, texture preservation, and warping were not achieved. However, the prototype effectively demonstrated the concept of clothing alignment and virtual garment placement.



Result

The experiment successfully produced a basic virtual try-on prototype capable of detecting body posture and overlaying a shirt on the detected torso region. The implementation provided practical understanding of virtual try-on workflows and garment alignment concepts.

Conclusion

This experiment demonstrated a simple implementation of a Virtual Try-On system using OpenCV and MediaPipe Pose. The developed prototype performed body detection, garment overlay, and manual alignment, helping in understanding the fundamental concepts involved in AI fitting systems. The experiment forms the basis for future enhancements such as garment warping, segmentation-based fitting, realistic cloth simulation, and advanced virtual try-on models.